

SUYOG KARKI

Data Science Intern Candidate | Machine Learning & Deep Learning

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PROFILE

Data Science student (BSIT), currently in my 4th year. I've worked through the full ML pipeline: cleaning data, EDA, feature engineering, model building, and tuning. I also recently finished deep learning (CNN, RNN, LSTM, transfer learning). Built 5 end-to-end ML projects, 3 deployed with Streamlit/Docker. Looking for a data science, data analyst, or ML internship where I can keep learning and contribute to real projects.

EDUCATION

Bachelor of Science in Information Technology (BSIT) - Data Science, 2023 – 2027
Kings College (Affiliated with Westcliff University, California) Kathmandu
Currently in 4th Year GPA: 3.54 / 4.0

Higher Secondary Education (+2) - Science, 2020 – 2022
Kathmandu Model College (KMC) Bagbazar
GPA: 3.35 / 4.0

TECHNICAL SKILLS

Languages

Python · SQL

Visualization

Matplotlib · Seaborn · Power BI

ML / Modeling

Scikit-learn · PyTorch · Optuna

Deep Learning

ANN · CNN · RNN · LSTM · Transfer Learning

Data

Pandas · NumPy · Excel

Deployment / Tools

Streamlit · Docker · Git & GitHub · PostgreSQL

Currently Learning

Computer Vision · NLP / Transformers

CERTIFICATIONS

- Supervised Learning with Scikit-learn | DataCamp | 2026
- Associate Data Scientist Certification | DataCamp | 2026
- SQL Associate | DataCamp | 2026

PROJECTS

Diabetic Retinopathy Severity Grading (APTOS 2019)

Python · PyTorch · torchvision · OpenCV · Scikit-learn · Pandas · NumPy

GitHub: <https://github.com/suyogkarki1/aptos-2019-blindness-detection.git> 📄

- Built a model that grades diabetic eye disease severity (0 to 4) on the APTOS 2019 dataset (3,662 retina images) with strong class imbalance (49% healthy vs 5% Severe).
- Used transfer learning with EfficientNet-B3 and framed the task as regression (one severity score with MSE loss) instead of 5-class classification, so near-miss grades are penalized less under quadratic weighted kappa.
- Found the model scored well locally but dropped on Kaggle's hidden test set, a sign of distribution shift, and fixed it by adding ~17,000 external retina images, keeping all diseased cases and sampling 8,000 healthy ones to reduce the class imbalance.

- Reached 0.915 kappa and 80% accuracy on a held-out test set, and improved the Kaggle private score from 0.848 to 0.894 (winner: 0.936).

Cardio — Heart Disease Risk Predictor

Python · Pandas · NumPy · Matplotlib · Seaborn · Scikit-learn · Streamlit

GitHub: <https://github.com/suyogkarki1/Clinical-Risk-Prediction-Diabetes-Heart-Disease> 

Live: https://clinical-risk-prediction-diabetes-heart-disease.streamlit.app/Heart_Disease 

- Built a heart disease risk predictor on the UCI Cleveland dataset (302 records), using a scikit-learn pipeline that imputes missing values, scales numeric features, and one-hot encodes categorical ones.
- Compared six models with 5-fold cross-validation and tuned each with GridSearchCV, keeping all preprocessing inside the folds to prevent data leakage, and handled class balance with class weights.
- Chose a tuned Random Forest for the best test accuracy and smallest train and test gap, reaching 0.86 ROC-AUC, 79% accuracy, 0.83 precision, and 0.76 recall on the test set.
- Deployed it as an interactive Streamlit web app with a live risk meter driven by the model's probability output.

Chini — Diabetes Risk Predictor for Women

Python · Pandas · NumPy · Matplotlib · Seaborn · Scikit-learn · SciPy · Streamlit · Docker

GitHub: https://github.com/suyogkarki1/diabetes_pipeline 

Live: <https://chinisugar.streamlit.app/> 

- Built a diabetes risk predictor for adult women on the Pima Indians dataset (768 records), using a scikit-learn pipeline that fills in missing values and scales features, so it works even with incomplete patient data.
- Ran Welch's t-tests comparing diabetic vs non-diabetic patients, using the t-statistic and p-value ($\alpha = 0.05$) to judge significance and Cohen's d for effect size, to find which measurements best separate the two groups.
- Chose Logistic Regression and scored on PR-AUC instead of accuracy because the data is imbalanced, reaching 0.81 ROC-AUC, 73% accuracy, and 0.72 recall on the test set.
- Deployed it as an interactive Streamlit web app, containerized with Docker for easy setup.

Customer Churn Prediction

Python · Pandas · Scikit-learn · Matplotlib · Seaborn

GitHub: <https://github.com/suyogkarki1/customer-churn-prediction> 

- Analyzed a Kaggle dataset of 10,000 European bank customers to identify behavioral patterns linked to churn, including credit score, account balance, and product usage habits.
- Achieved 86.35% accuracy and 0.87 ROC-AUC using Gradient Boosting Classifier, selected after benchmarking against simpler models including Logistic Regression and Decision Tree.
- Feature importance analysis revealed Age and NumOfProducts as the strongest churn predictors, suggesting retention teams should prioritize older single product customers as the highest risk segment.

Kathmandu Rain Prediction

Python · Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn · Streamlit

GitHub: <https://github.com/suyogkarki1/Rain-Prediction-> 

Live: <https://barshamausam.streamlit.app/> 

- Sourced a 10 year Kathmandu weather dataset (2015 to 2025, 3,468 records) from Kaggle, and performed data cleaning, unit conversion, and feature engineering across 14 meteorological variables.
- Built and evaluated four classification models (Logistic Regression, Decision Tree, K-Nearest Neighbors, and Random Forest) using stratified 6-fold cross-validation; selected Random Forest, achieving 84% accuracy and 0.90 ROC-AUC on held-out test data.
- Applied bootstrap resampling to estimate 2026 monthly rain probabilities with lower and upper uncertainty bounds, and developed a separate SARIMA seasonal time series model to forecast monthly rainy day percentages for comparison.

- Deployed the trained model as an interactive Streamlit web application (SkyCast) enabling real time rain classification from user defined weather inputs and visualizing seasonal forecast trends.

UAE ID Card Number Extraction Pipeline

Python · YOLOv11 (Ultralytics) · OpenCV · Tesseract OCR · PostgreSQL

GitHub:https://github.com/suyogkarki1/object_detection 

- Annotated a custom dataset of 424 UAE ID card images on Ultralytics HUB, labeling the ID number field as a dedicated detection class, split into 405 train / 10 validation / 9 test images.
- Trained and deployed a custom YOLOv11 model via Ultralytics' hosted endpoint, achieving 99.5% mAP50 and 95.0% recall on ID number localization.
- Built an OCR pipeline (OpenCV preprocessing + Tesseract) to extract and clean detected ID numbers via regex, then persisted records to PostgreSQL with conflict safe deduplication.

LANGUAGES

Nepali

Native

English

Fluent